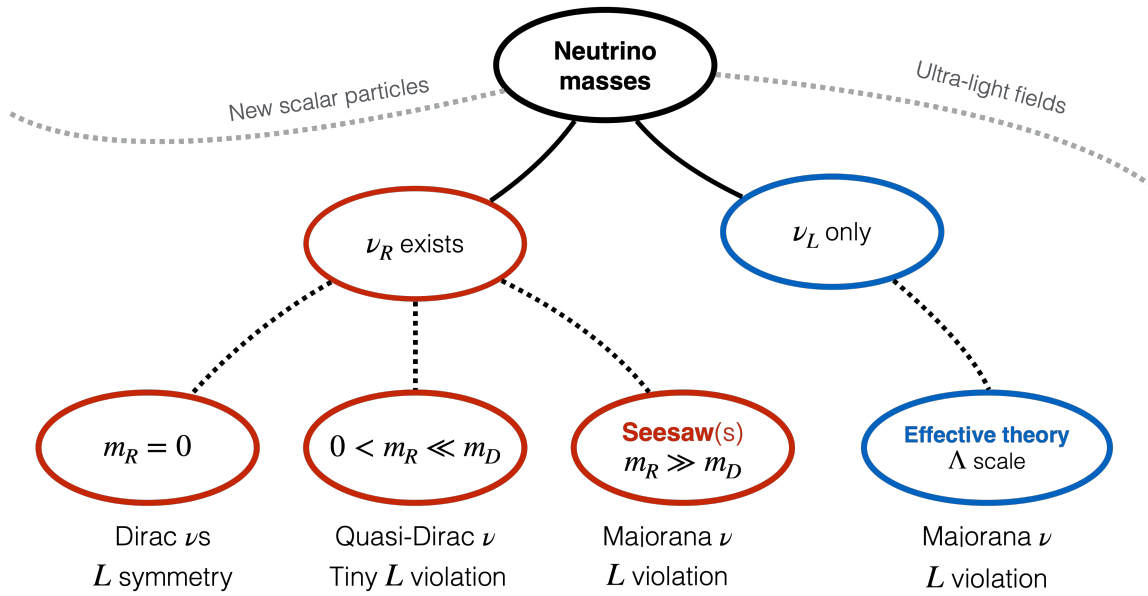


Neutrino Mass Mechanisms

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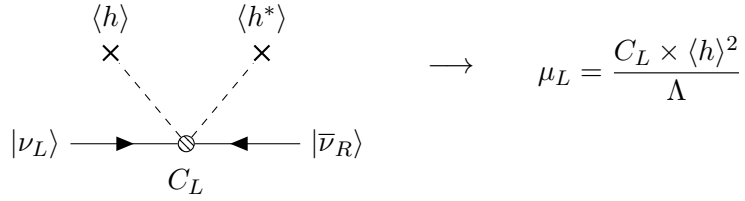


1 Paths to Neutrino Masses

We will now turn to the different mechanisms by which neutrinos can acquire a mass in minimal extensions of the SM.

Majorana neutrinos The first case to consider is the one where the SM field content is left unchanged. In this case, only left-chiral neutrino fields exist, $\nu_L(x)$, and therefore, the only possible mass term for neutrinos is of Majorana type. As we argue below, this mass term by itself is not gauge invariant, so we must give something up; that something will be renormalizability.

The following diagram shows how a SM neutrino Majorana mass term, μ_L , could arise in the SM:



Here I indicated that the Majorana mass μ_L depends on C_L , a dimensionless constant, and Λ , some new energy scale we discuss shortly. This is similar to the Higgs mechanism for Dirac fermion masses, but with an important distinction: we needed not one, but two(!) Higgs VEV insertions. A quick inspection of the balance of hypercharges shows why:

$$Y(\bar{\nu}_R) = -Y(\nu_L) = +1 \text{ and } Y(h) = +1. \quad (1.1)$$

Note that I used the vev of the complex-conjugate of the Higgs field, h^* , which has $Y_W(h^*) = -Y_W(h)$. This distinction is immaterial because $\langle h \rangle$ is just a constant, but I wanted to draw attention to the fact that in the way the diagram is drawn, you need an anti-Higgs field to balance things out.

The Majorana mass term in the Lagrangian requires then two Higgs fields. The lowest dimensional operator that can generate such a mass term is dimension-5 *Weinberg operator* (check this is indeed gauge invariant):

$$\mathcal{L} \supset -\frac{C_L}{\Lambda} (\bar{L}\tilde{H})(\tilde{H}^T L^c) \xrightarrow{\text{EWSB}} -\frac{1}{2}\mu_L \bar{\nu}_L^c \nu_L + \dots, \quad \mu_L = \frac{C_L \langle h \rangle^2}{\Lambda}, \quad (1.2)$$

where L is the lepton doublet, H is the Higgs doublet,

$$\tilde{H} = \epsilon H^* = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} H^+ \\ H^0 \end{pmatrix}^* \xrightarrow{\text{EWSB}} \begin{pmatrix} \langle h \rangle \\ 0 \end{pmatrix} \quad (1.3)$$

is the conjugate Higgs doublet (ϵ is the antisymmetric tensor), and $\nu^c = C\bar{\nu}^T$ is the charge-conjugate of the neutrino field.



Because this is a higher-dimensional operator, it is non-renormalizable, and we must accept that at the scale Λ/C_L , some additional fields/physics must exist to complete the theory. We will see some examples later on, but there are many. The power of this effective description is precisely that it is so general. If neutrinos are Majorana particles, this framework *describes* (even if it does not fully *explain*) all of the physics of neutrino masses without committing to any particular model.

In the realistic three-neutrino generation case, there must be at least two non-zero neutrino masses to explain the observed oscillation data, and so at the very least two different Weinberg operators. Because of flavor mixing, this number is actually higher. In that case, we can write the coefficient C_L as a 3×3 matrix in flavor space, C_L^{ij} , where $i, j = e, \mu, \tau$ are the lepton flavors:

$$\mathcal{L}_{\text{Weinberg}} = -\frac{1}{\Lambda} C_L^{ij} (\bar{L}_i \tilde{H})(\tilde{H}^T L_j^c) \xrightarrow{\text{EWSB}} -\frac{1}{2} \mu_L^{ij} \bar{\nu}_{Li}^c \nu_{Lj} + \dots, \quad \mu_L^{ij} = \frac{C_L^{ij} \langle h \rangle^2}{\Lambda}. \quad (1.4)$$

Diagonalizing these mass terms gives us the neutrino mass eigenstates and the neutrino mixing matrix (not yet the PMNS, as we have accounted for any rotation of the charged lepton sector).

What should the scale Λ be? We can estimate it by using the observed neutrino mass scale, $\mu_L = m_\nu \sim 0.1$ eV, and the Higgs vev, $\langle h \rangle \sim 246$ GeV:

$$\Lambda \sim \frac{C_L \langle h \rangle^2}{m_\nu} \sim 10^{14} \text{ GeV} \times C_L. \quad (1.5)$$

This tells us that new physics can appear anywhere up to this scale (it may be that the coefficient $C_L \ll 1$).

Dirac neutrinos? If the SM is extended to include RH neutrinos ν_R , then neutrinos can be Dirac particles. The Lagrangian would look like

$$\mathcal{L} \supset -y_\nu \bar{L} \tilde{H} \nu_R + \text{h.c.} \xrightarrow{\text{EWSB}} -m_D \bar{\nu}_L \nu_R + \dots, \quad m_D = \frac{y_\nu \langle h \rangle}{\sqrt{2}}. \quad (1.6)$$

Taking $y_\nu \sim 10^{-12}$ gives us the approximate neutrino mass scale, $m_D = m_\nu \sim 0.1$ eV. This would mean neutrinos get a mass in the same way as all other SM fermions, but with a very small Yukawa coupling (you may not be too bothered about this, after all $y_e \sim 10^{-6}$ is also pretty small).

The catch, however, is that right-chiral neutrinos can also have a Majorana mass term of their own. This time, however, without the need for the Higgs to be involved at all. Right-chiral neutrinos do not carry any SM charge (they are sterile, singlets under all SM gauge symmetries), so their Majorana mass is not subject to the gauge-invariance constraint. We do not need to go to higher-dimensional operators like in Eq. (1.4). If this happens, neutrinos are **not** purely Dirac particles. We will explore the different limits of this scenario in Section 2.

Diagrammatically, the right-chiral Majorana mass term looks like:

$$|\nu_R\rangle \longrightarrow \times \longleftarrow |\bar{\nu}_L\rangle$$

μ_R

We now use μ_R to differentiate this mass from μ_L above. While μ_L is tied to the Higgs VEV, μ_R is a completely free parameter. If $\mu_R = 0$, then the RH neutrino is a Dirac partner to the LH neutrino, and we recover the Dirac neutrino case above.



To summarize: we have seen two paths to extend the SM to endow neutrinos with a mass.

1. Introduce a Majorana mass term for the left-handed neutrinos ν_L with a double Higgs VEV insertion.
2. Introduce a new field ν_R to make neutrinos Dirac particles.

In each one of these cases, we have important consequences: for (1), it implies new physics at some scale Λ , and for (2), it implies the existence of a new particle, ν_R , that we have never seen, and that may admit its own arbitrary Majorana mass μ_R , possibly spoiling the Dirac nature of the neutrino.

Lepton number. Before moving on, let me say a few words about lepton number L . As the name suggests, in the SM, we assign

$$L(\nu_L) = +1, \quad L(\bar{\nu}_R) = -1, \quad L(e_L) = +1, \quad L(\bar{e}_R) = -1. \quad (1.7)$$

All other SM particles that are not leptons do not carry lepton number, $L(\text{quarks}) = L(\text{mediators}) = L(\text{higgs}) = 0$.

The SM happens to conserve lepton number in all of its interactions and masses. There simply is no parameter in the SM that directly violates lepton number. This is an accidental symmetry: it was never imposed, but it just so happens to be true. If neutrinos are Majorana, however, it is then violated by $\Delta L = 2$ units.

This is true for both the left-handed *and right-handed* Majorana mass terms, namely μ_L (from the Weinberg operator) and μ_R , respectively. After all, right-chiral neutrinos are still leptons,

$$L(\nu_R) = +1, \quad L(\bar{\nu}_L) = -1. \quad (1.8)$$

2 Single Generation Neutrino Mass Matrix with ν_R

For a single generation of neutrinos (think $\nu_L = \nu_e$, for instance), the most general mass matrix that we can write in our Lagrangian is

$$\begin{aligned}\mathcal{L} \supset & -m_D \bar{\nu}_L \nu_R - m_D \bar{\nu}_R \nu_L - \frac{1}{2} \mu_L \bar{\nu}_L \nu_L - \frac{1}{2} \mu_R \bar{\nu}_R \nu_R + \text{h.c.} \\ & = -\frac{1}{2} \begin{pmatrix} \bar{\nu}_L & \bar{\nu}_R^c \end{pmatrix} \begin{pmatrix} \mu_L & m_D \\ m_D & \mu_R \end{pmatrix} \begin{pmatrix} \nu_L^c \\ \nu_R \end{pmatrix} + \text{h.c.}\end{aligned}\tag{2.1}$$

where $|\nu_L\rangle$ is the usual left-handed neutrino and $|\nu_R\rangle$ is a new right-handed neutrino (non-interacting). Recall that CPT invariance means ν_L and ν_R must come with their respective antiparticles, $\bar{\nu}_L$ and $\bar{\nu}_R$. Here, m_D is a Dirac mass term that can come from the Higgs mechanism, μ_L is a Majorana mass term for the Weakly-interacting neutrino (à la Weinberg operator), and μ_R is a Majorana mass term for the sterile neutrino.



Find the rotation angle θ that diagonalizes this matrix. Now find the eigenvalues of this matrix. Expand them in the following two limits:

1. $\mu_L, \mu_R = 0$ (the Dirac limit)
2. $\mu_L, \mu_R \ll m_D$ (the pseudo-Dirac/quasi-Dirac limit)
3. $\mu_R \gg m_D$ (the seesaw limit, where all particles are of the Majorana type). You can either keep μ_L general (assuming $\mu_R \gg \mu_L$) or set it to zero.

Using the very useful formula for diagonalizing a 2×2 symmetric matrix, we get

$$\tan 2\theta = \frac{2m_D}{\mu_R - \mu_L}.\tag{2.2}$$

Note that this formula is valid for any combination of real parameters. The point here is that the flavor basis (the weak-interaction eigenstates) is rotated by an angle θ to get to the mass basis (the propagating eigenstates):

$$D = \text{diag}(m_1, m_2) = R(\theta)^T \cdot M \cdot R(\theta) \text{ with } R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.\tag{2.3}$$



To find the eigenvalues of the mass matrix, follow a few different procedures. You can perform the rotation above by hand, or you can solve the characteristic polynomial $\det(M - \lambda I) = 0$, or you can use the fact that for a 2×2 matrix, the eigenvalues are given by

$$m_{1,2} = \frac{1}{2} \left(\text{Tr}(M) \pm \sqrt{\text{Tr}(M)^2 - 4 \det(M)} \right). \quad (2.4)$$

In our case, we have $\text{Tr}(M) = \mu_L + \mu_R$ and $\det(M) = \mu_L \mu_R - m_D^2$. Therefore, the eigenvalues are

$$m_{1,2} = \frac{1}{2} \left(\mu_L + \mu_R \pm \sqrt{(\mu_L - \mu_R)^2 + 4m_D^2} \right). \quad (2.5)$$

This relation comes from the fact that

$$\text{Tr}(M) = \text{Tr}(R^T D R) = \text{Tr}(D R R^T) = \text{Tr}(D) = m_1 + m_2, \quad (2.6)$$

and

$$\det(M) = \det(R^T D R) = \det(R^T) \det(D) \det(R) = \det(D) \det(R^T R) = m_1 m_2. \quad (2.7)$$

These are two equations with two unknowns, so you can solve for $m_{1,2}$ by solving the underlying quadratic equation. This is analogous to solving the characteristic polynomial, but it can be faster. For instance, for a zero-determinant matrix, you immediately know one eigenvalue is zero, so you can get the other one from the trace.

Now, let's define the following useful parameters:

$$\mu \equiv \mu_R + \mu_L, \quad \Delta\mu \equiv \mu_R - \mu_L. \quad (2.8)$$

In terms of these, the eigenvalues can be written as

$$m_{1,2} = \frac{1}{2} \left(\mu \pm \sqrt{\Delta\mu^2 + 4m_D^2} \right). \quad (2.9)$$

Now, we can expand these eigenvalues in the different limits.

Dirac In the Dirac limit, $\mu_L, \mu_R = 0$, we have $\mu = 0$ and $\Delta\mu = 0$, so

$$\boxed{m_{1,2} = \pm m_D}. \quad (2.10)$$

This is saying that these two neutrino states combine into a Dirac fermion of mass m_D .

i

A negative mass fermion?! Not to worry, this is just telling us that a Dirac fermion is made up of two degenerate Majorana fermions with the same mass, but opposite CP phases. You can see that by writing the mass terms for the mass eigenstates in the Lagrangian:

$$\mathcal{L} \supset -\frac{1}{2} (m_1 \nu_1^T C \nu_1 + m_2 \nu_2^T C \nu_2) + \text{h.c.} = -\frac{1}{2} m_D (\nu_1^T C \nu_1 - \nu_2^T C \nu_2) + \text{h.c.}$$

Recall that a Majorana mass term is of the form $\psi^T C \psi = \bar{\psi} \psi^c$ where $\psi^c = C \bar{\psi}^T$. Here ψ is a 4-component spinor and C is the charge-conjugation matrix (in the Dirac basis, $C = i\gamma^2 \gamma^0$). Now, defining $\nu = (\nu_1 + i\nu_2)/\sqrt{2}$ and $\bar{\nu} = (\nu_1 - i\nu_2)/\sqrt{2}$, we get

$$\mathcal{L} \supset -m_D \bar{\nu} \nu,$$

which is the usual Dirac mass term. Note that we are free to redefine the phases of the fields to make masses positive. For example, we could redefine $\nu_2 \rightarrow i\nu_2$ to make both masses positive. This means that the phase of ν_2 is different from that of ν_1 by $\pi/2$, which is what gives us the opposite CP phases.

i

The opposite CP phase is meaningful experimentally as well. For those familiar with neutrinoless double-beta decay, you can spot something interesting here. For this single generation toy example with Dirac neutrinos, we get

$$m_{\beta\beta} \sim ||U_{e1}^2 m_1 + U_{e2}^2 m_2|| = ||\frac{1}{2} m_D - \frac{1}{2} m_D|| = 0,$$

where we used the fact that the rotation matrix is $U = R(\pi/4)$ in the Dirac limit, so $U_{e1} = U_{e2} = 1/\sqrt{2}$. The relative minus sign comes either from keeping the negative mass for m_2 or from the CP phase if we redefine $\nu_2 \rightarrow i\nu_2$.

Quasi-Dirac In the quasi-Dirac limit, $\mu_L, \mu_R \ll m_D$, we have $\mu \ll m_D$ and $\Delta\mu \ll m_D$. Let us rewrite the general expression for the eigenvalues as

$$m_{1,2} \simeq m_D \left(\frac{\mu}{2m_D} \pm \sqrt{1 + \left(\frac{\Delta\mu}{2m_D} \right)^2} \right). \quad (2.11)$$

We now treat $\mu/(2m_D)$ and $\Delta\mu/(2m_D)$ as small parameters and expand the square root to first order using $\sqrt{1+x} \approx 1 + x/2$ for $x \ll 1$.

$$m_{1,2} \approx m_D \left(\frac{\mu}{2m_D} \pm \left(1 + \frac{1}{2} \left(\frac{\Delta\mu}{2m_D} \right)^2 \right) \right) = \pm m_D + \frac{\mu}{2} \pm \frac{(\Delta\mu)^2}{8m_D}. \quad (2.12)$$

In reality, the last term is negligible compared to the first two, so we can write (suggestively)

$$\boxed{m_1 \simeq -\left(m_D - \frac{\mu}{2}\right), \quad m_2 \simeq +\left(m_D + \frac{\mu}{2}\right)}. \quad (2.13)$$

Recalling that the overall minus sign is just a phase, this expression is telling us that we have two *nearly-degenerate* Majorana neutrinos with masses close to m_D , split by a total amount μ .

Seesaw In the seesaw limit, $\mu_R \gg m_D$. Initially, let us set $\mu_L = 0$ for simplicity. In that case, $\mu = \mu_R$ and $\Delta\mu = \mu_R$ and the eigenvalues become

$$m_{1,2} = \frac{\mu_R}{2} \pm \frac{1}{2} \sqrt{\mu_R^2 + 4m_D^2}. \quad (2.14)$$

Expanding in the small parameter m_D/μ_R , we get

$$m_1 \simeq -\frac{m_D^2}{\mu_R}, \quad m_2 \simeq \mu_R. \quad (2.15)$$

This is the seesaw mechanism at work: one neutrino becomes very heavy, with mass $\sim \mu_R$, while the other becomes very light, with mass $\sim m_D^2/\mu_R$.

Reintroducing μ_L does not change the qualitative picture, but it does modify the light neutrino mass to $m_1 \simeq \mu_L - m_D^2/\mu_R$. Note that in any of the scenarios we discussed here we can obtain neutrino masses without resorting to the Weinberg operator, so there is no reason to keep μ_L in the game.

?

In the seesaw limit, call the physical basis states ν_ℓ and ν_h , where ν_h is the heavy neutrino, called a heavy neutral lepton (HNL). What is the relation between the weak-interaction eigenstate ν and these mass eigenstates in terms of θ ? Which state ν_ℓ or ν_h will interact more strongly with the rest of the SM particles?

Using the rotation in Eq. (2.3), we have

$$|\nu_L\rangle = \cos\theta |\nu_\ell\rangle + \sin\theta |\nu_h\rangle, \quad |\nu_R\rangle = -\sin\theta |\nu_\ell\rangle + \cos\theta |\nu_h\rangle. \quad (2.16)$$

The mixing angle in the seesaw limit is small, so using Eq. (2.2) we get

$$\theta \simeq \frac{m_D}{\mu_R}, \quad (2.17)$$

so that we can write

$$|\nu_L\rangle \simeq |\nu_\ell\rangle + \theta |\nu_h\rangle, \quad |\bar{\nu}_R\rangle \simeq |\nu_h\rangle - \theta |\nu_\ell\rangle. \quad (2.18)$$

The light mass state is the one that most closely resembles ν_L , and therefore, is the one that interacts more with SM particles.

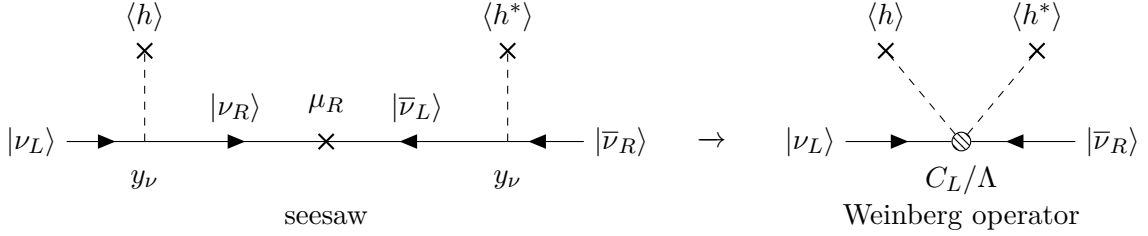
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Identify ν_ℓ state with the neutrinos we observe in the laboratory and take $m_{\nu_\ell} \sim 0.05$ eV. Recalling that m_D comes from the Higgs mechanism, $m_D = yv/\sqrt{2}$ with $v = 246$ GeV and y a coupling constant, we will take m_D to be in the range $[m_e, m_t] = [511 \text{ keV}, 176 \text{ GeV}]$, corresponding to $y \sim [3 \times 10^{-6}, 1]$. What are the values of μ that you need in order to obtain the assumed value for m_{ν_ℓ} ? What values of the mixing angle do these correspond to?

Putting in the numbers, we get roughly, $\mu_R \sim [10^3 \text{ GeV}, 10^{14} \text{ GeV}]$ and $\theta \sim [10^{-7}, 10^{-12}]$. You can see that even if the Dirac mass term is as small as the electron one, the required Majorana mass (and therefore the mass of the propagating state ν_h) will be very large and will be quickly untestable as you raise m_D . All relevant cross sections and decay rates for detecting ν_h , in fact, scale as θ^2 , so that makes these particles very hard to observe in the laboratory, even if we had the energy to produce them. We say then, that ν_h has *weaker-than-Weak* interactions, meaning that the only part of ν_h that does interact (a fraction θ of its superposition) interacts only through the Weak force.

3 Reinterpreting the Weinberg Operator

The seesaw mechanism is an example of what kind of new physics can “UV complete” the Weinberg operator (it makes the theory renormalizable and well behaved in the UV/high energies). We can pretty easily see how that comes about diagrammatically:



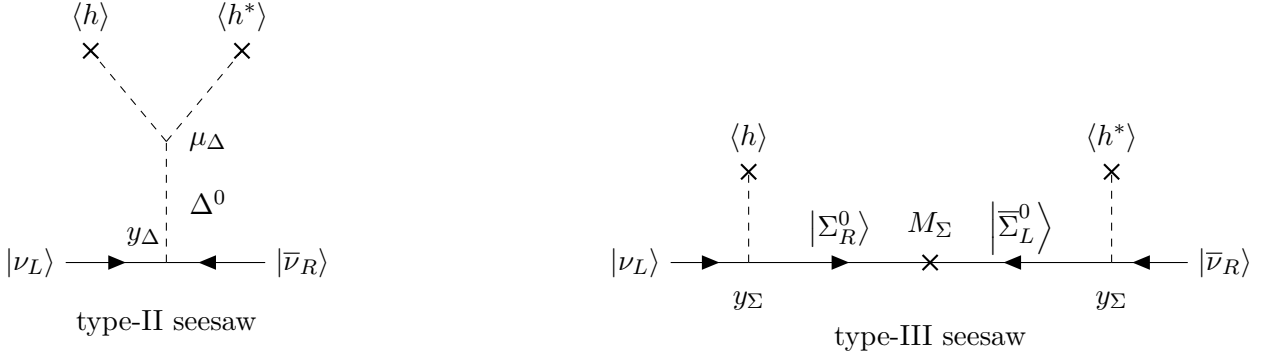
Because the heavy neutrino ν_h is very heavy, the middle propagator suppresses the amplitude for this process, and we can shrink it to a contact interaction at low energies (much in the same way we did with the W boson when using G_F):

$$\frac{y_\nu \langle h \rangle}{\sqrt{2}} \times \frac{1}{\mu_R} \times \frac{y_\nu \langle h \rangle}{\sqrt{2}} = \frac{y_\nu^2 \langle h \rangle^2}{2\mu_R} \longrightarrow \frac{C_L}{\Lambda} \langle h \rangle^2, \quad (3.1)$$

As long as our experiment/measurement is not energetic enough to produce $\nu_h \sim \nu_R$ on shell, we can treat it as a heavy virtual particle (in the same way we treat the W boson when using G_F).

This UV completion of the Weinberg is called the *type-I seesaw mechanism*. It is not unique, but it is the simplest and most popular one. The other types of seesaw mechanisms are called type-II and type-III, and they involve adding new particles to the SM that are not singlets under the SM gauge group. For the Type-II, one adds a scalar triplet $\Delta = (\Delta^{++}, \Delta^+, \Delta^0)$, and for the Type-III, one adds a fermion triplet $\Sigma = (\Sigma^+, \Sigma^0, \Sigma^-)$. These models, like the Type-I seesaw, also generate the Weinberg operator at low energies, but they come with additional particles (the charged particles in the triplets) that can be searched for at high energies. Contrary to the Type-I seesaw, these models require the masses of the new particles to be heavy, as otherwise the electrically-charged states would already have been observed. Because they are charged under the weak interactions, they also lead to additional contributions to lepton flavor violation, electroweak precision observables, and other low-energy processes, which can be used to constrain these models.

The schematic diagrams for type-II and type-III seesaw mechanisms look like



Each one of these models must have a source of lepton number violation that will eventually get packaged into the Weinberg operator at low energies. For the Type-I seesaw, that was just the Majorana mass of ν_R . For the Type-II seesaw, that is the μ_Δ term, which is a trilinear coupling between the Higgs and the scalar triplet Δ :

$$\mathcal{L} \supset -\mu_\Delta H^T \epsilon \Delta^\dagger H + \text{h.c.}, \quad \Delta = \begin{pmatrix} \Delta^+/\sqrt{2} & \Delta^{++} \\ \Delta^0 & -\Delta^+/\sqrt{2} \end{pmatrix} \quad (3.2)$$

Finally, for the Type-III seesaw, that is the Majorana mass of the neutral component of the fermion triplet $\Sigma = (\Sigma^+, \Sigma^0, \Sigma^-)$:

$$\mathcal{L} \supset -\frac{1}{2} M_\Sigma \text{Tr}(\bar{\Sigma}^0 \Sigma^0) + \text{h.c.}, \quad \Sigma = \begin{pmatrix} \Sigma^+/\sqrt{2} & \Sigma^{++} \\ \Sigma^0 & -\Sigma^+/\sqrt{2} \end{pmatrix}. \quad (3.3)$$

These three seesaw mechanisms provide *all* the possible ways to UV complete the Weinberg operator at tree level. There are a huge number of possibilities once we are allowed to go to higher loops. A few classic examples are the Zee model, the Zee-Babu model, and the scotogenic model, many of which contain heavy neutral particles that can be provide a dark matter candidate.

4 Low Scale Seesaw Mechanisms

The right-handed neutrino seesaw is special in that it allows for the possibility of a low-scale seesaw mechanism, where the Majorana mass of the right-handed neutrino is not too far from the electroweak scale. Usually these models are hard to test because the mixing angle θ is very small, so the heavy neutral lepton ν_h has very weak interactions with the SM. However, there are some models that allow for a low-scale seesaw mechanism with a large mixing angles.

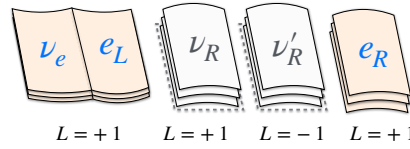
These are called inverse, linear, and extended seesaw mechanisms. **(See problem set for more details.)**

Example: The *inverse* seesaw

Interaction (flavor) basis

$$(\nu_L \quad \nu_R \quad \nu'_R) \begin{pmatrix} 0 & m_D & 0 \\ m_D & 0 & M \\ 0 & M & m_R \end{pmatrix} \begin{pmatrix} \nu_L \\ \nu_R \\ \nu'_R \end{pmatrix}$$

Two kinds of sterile neutrino per active neutrino ν_α



$$m_1 = \frac{m_D^2}{M^2} m_R \text{ (Majorana particle) and } m_{2,3} \sim M \pm \frac{m_R}{2} \text{ (quasi-Dirac pair).}$$

Neutrino masses are small due to quasi-preservation of lepton number.

Bonus: the mixing $\theta \sim \frac{m_D}{M}$ can be large when M is within experimental reach.



M. Hostert

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Figure 1: Schematic diagram of the inverse seesaw mechanism. The smallness of the light neutrino mass is controlled by the small Majorana mass m_R of the sterile neutrino ν'_R .

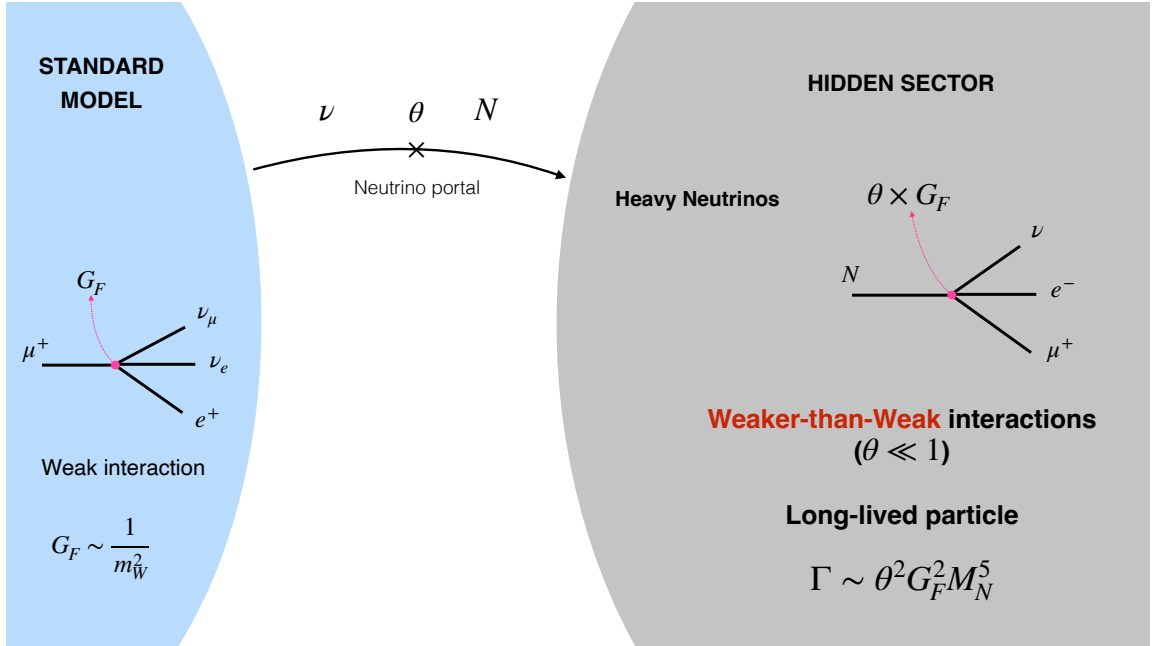
5 Phenomenology of Heavy Neutral Leptons

Depending on their mass, the heavy neutral leptons can be searched for in a variety of ways: through oscillations (3+1 neutrino oscillations at short baselines), beta decay kink searches, peak searches in meson decays (e.g., $K^+ \rightarrow e^+ \nu_h$), beam dump experiments, and collider searches. The key is that the heavy neutral leptons ν_h interactions with the SM are suppressed by the small mixing angle θ , so we say they are “weaker-than-Weak”. This applies to both their production and decay rates. In the eV range, they are too light to decay but would be too hot to be dark matter. In the keV range they are a good warm dark matter candidate, though these scenarios usually require alternative cosmologies or additional new physics. In the MeV to GeV range, they are long-lived and can be searched for in beam dump experiments, meson decays, and colliders. At the LHC energy scale, HNLs can be produced in W and Z decays, and they can be searched for in displaced vertex searches.

In the MeV-GeV mass range, we can estimate the lifetime of an HNL by scaling the muon lifetime:

$$\tau_{\nu_h} \sim \frac{\tau_\mu}{\theta^2} \left(\frac{m_\mu}{m_{\nu_h}} \right)^5. \quad (5.1)$$

For instance, if we take $m_{\nu_h} \sim 100$ MeV and $\theta^2 \sim 10^{-6}$, the lifetime of the HNL would be $\tau_{\nu_h} \sim 1$ s, which is much longer than the muon lifetime of $\tau_\mu \sim 10^{-6}$ s. See also the HNL problem set.



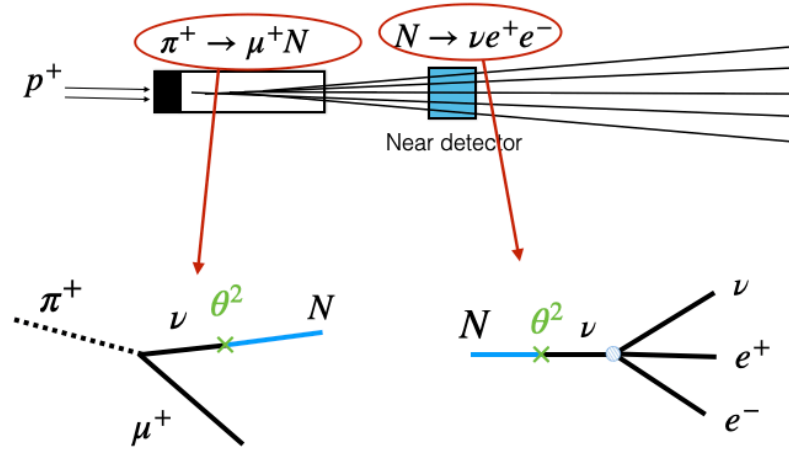


Figure 2: Production and decay of a heavy neutral lepton in a neutrino experiment.

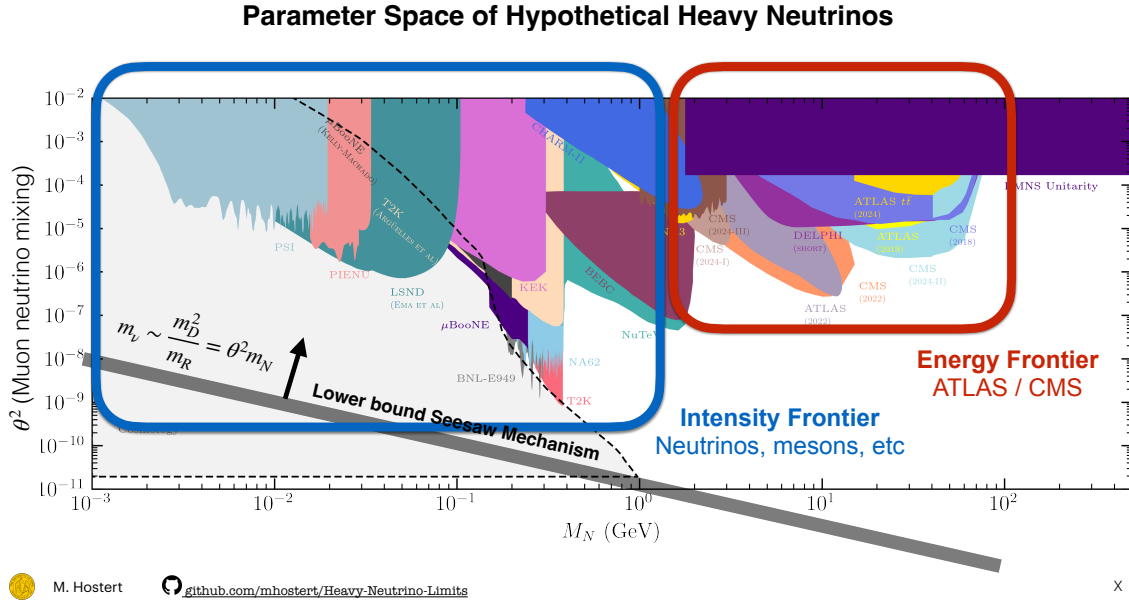


Figure 3: Parameter space of a heavy neutral lepton that mixes exclusively with the muon neutrino.

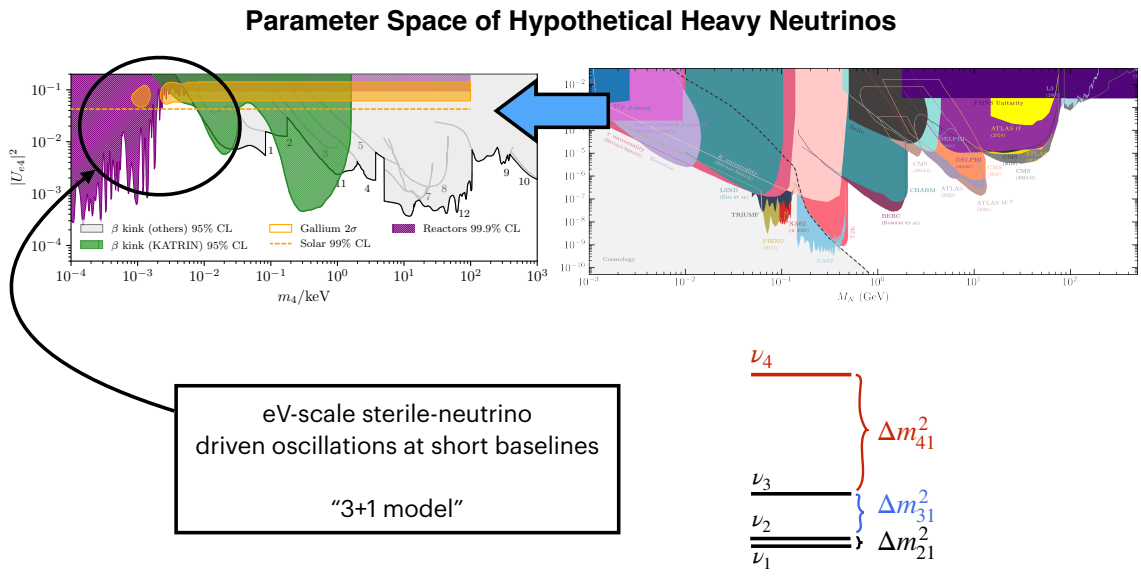


Figure 4: The light sterile neutrino parameter space, showing the mass region relevant for eV-scale sterile neutrinos (mixing exclusively with the electron flavor now).